

Racial Segregation and Black-White Longevity Disparities

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1 Abstract

Residential, racial segregation defines the life chances of Black Americans and White Americans, but in potentially different ways. Despite implicit assumptions that segregation advantages White Americans, little empirical research has investigated whether this is the case. This empirical gap limits our theoretical understanding of the causes of pronounced White resistance to residential integration, despite the fact that desegregation in educational and institutional settings had virtually no impact on Whites' outcomes. While past work has attributed White reactions to integration to perceived loss of status position in the racial hierarchy, explanations are incomplete without understanding if segregation advantages Whites or whether the costs are primarily psychological. To provide clarity about the causes of resistance to integration, I study the effects of segregation on the longevity of Black and White Americans using data from the 1940 Full Count Census linked to Numident death records and plausibly exogenous variation in the density and Federal revenue share of local governments. I show that racial segregation reduces the longevity of Black and White Americans, albeit at different magnitudes. White aversion to integration — rather than advantaging Whites — actually harms their health. Rather than resulting in a zero-sum perspective, policies promoting residential integration would substantially improve Black and White health.

2 Introduction

Black-White racial segregation is a defining feature of American cities both because of its persistence across space and time but also because of its exceptionally high levels in comparison to the segregation of other ethnoracial groups (Elbers 2021; Frey and Farley 1996). Historically, segregation peaked in the 1960s and 1970s, particularly in the North following the Great Migration of African Americans from the South to new destinations (Logan and Stults 2022; ?). In racially segregated places, Black-White disparities in health, longevity, and socioeconomic status are persistently wide (Harper, MacLehose, and Kaufman 2014), despite segregation declining since the passage of the Civil Rights act of 1964.

Segregation, an overt manifestation of racial inequality, is often thought to profoundly shape the contexts of exposure to and residential environments of Black and White Americans in fundamental, but in potentially countervailing ways. Segregation leads to concentrated exposure to poverty, violent crime, and pollution in Black neighborhoods in ways that impact their health and wellbeing. The racially uneven distribution or risk and disadvantage is thought to positively advantage White Americans by reducing their exposures to these same adverse consequences in ways that are health advantaging (Massey and Denton 2003); Light and Thomas (2019); Wodtke et al. (2023)]. In the context of pronounced White resistance to racial integration over time, in the form of overt racial backlash, White flight, and avoidance from diversity (Crowder and South 2008; Rich n.d.), sociological explanations have primarily attributed these reactions to White anxiety about losing status position in the racial hierarchy (Bobo and Zubrinsky 1996; Wetts and Willer 2018). Yet, empirical research on the effects of segregation on White Americans and whether segregation advantages the well-being of Whites has received less empirical attention.

Two theoretical perspectives provide interpretations of Whites’ resistance to segregation. The **Zero-Sum Hypothesis** argues that Whites’ resist integration because it comes at the expense real advantages afforded by racial inequality (McGhee 2021; Wetts and Willer 2018). In contrast to this perspective, a Du Boisian **Wages of Whiteness Hypothesis** argues that Whites do not actually benefit from racial inequality but are paid a “psychological wage” of from Whiteness, despite the fact that they are actually unaffected or in some cases similarly harmed (DuBois 1935:1935: 700–701). Past studies show that in voting patterns and public opinion polls, Whites’ oppose reform that would improve the conditions of Black Americans, such as welfare and redistributive policies (Gilens 2000; McGhee 2021; Wetts and Willer 2018) aligning with the belief in the **Zero-Sum Hypothesis**. In contrast, empirical studies of the effects of school desegregation on White Americans show that these policies did not affect White Americans’ outcomes positively or negatively despite improving outcomes for Black Americans (Anstreicher, Fletcher, and Thompson n.d.; Johnson 2019) providing evidence for the **Wages of Whiteness Hypothesis**. Whether benefits of segregation for Whites are real or perceived is currently unclear due to the lack of empirical research on the effects of segregation on Whites.

In this paper, I study the effects of racial segregation on the longevity of Black and White Americans born in the early 20th Century (1905-1920), using data from the 1940 Census linked to CenSoc Numident death records ($N = 2,157,524$) and an instrumental variable strategy leveraging variation in the density of governments and their revenue share in U.S. counties to instrument for racial segregation (Cutler and Glaeser 1997). I study longevity as a measure of life chances because it is both an important dimension of population health and an axis of social inequality between Black and White Americans (Wrigley-Field 2024); Maralani and Portier (2021)]. Beyond health status, longevity disparities have far-reaching implications for social networks, collective memory, culture, political participation, and social movements (Wrigley-Field

2025) Longevity thus is an omnibus measure of non-economic wellbeing and an important descriptor of the summation of life chances (Montez et al. 2020; ?).

Results indicate that segregation decreases Black longevity by **-2.8 years** and decreases White longevity by **-.8 years years overall**. The negative effects of segregation for Black Americans are especially pronounced in the South (-4 years), whereas effects are negative but insignificant outside of the South. These results for Black Americans are consistent with literature on the effects of the Great Migration suggesting unclear or even negative effects of leaving the South (Black et al. 2015). For White Americans, the effects of segregation within and outside the south are similar in magnitude and negative. These results emphasize the importance of the effects of structural racism on the health of both Black and White Americans, that while segregation harms the longevity of Black Americans, it also harms the health of White Americans. Consistent with the **Wages of Whiteness Hypothesis**, White aversion to integration — rather than advantaging Whites — actually harms their health. Policies promoting integration would substantially improve Black health outcomes while also benefiting White Americans. In the context of a Black-White life expectancy gap of 6.3 years in 2000 (Arias n.d.), these results imply that moving from a perfectly segregated to perfectly integrated county would narrow the gap to 3.7 years $((77.4 + .8) - (71.7 + 2.8)) = 3.7$ or about half of the observed gap.

3 Background

3.1 White Resistance to Integration and the Consequences of Segregation for White Americans

Since the passage of the Civil Rights Act of 1964, which outlawed legally codified segregation, de facto residential segregation has emerged as the “‘structural linchpin’ of race relations in America,” (Bobo and Zubrinsky 1996 : 883) and represents one of the most consequential and visible dimensions of Black-White

inequality in the U.S. Sociological research has long documented the significance of the differential residential arrangements of Black and White Americans, beginning with work by W.E.B. Du Bois in *The Philadelphia Negro* that highlighted how residential context engendered poor educational, economic, and health outcomes for Black residents (?). Thus, the residential separation between Black and White Americans is thought to be a physical manifestation of racial inequality and a driving force behind persistent Black-White inequalities both within and across generations (?; ?). The unequal distribution of disadvantage in segregated cities continue to have implications for the concentration of poverty in Black neighborhoods ([Massey and Denton 2003](#)), exposure to violence ([Light and Thomas 2019](#)), and decreased economic and educational opportunities (?). Cumulatively, these factors play a role in shaping profound Black-White health disparities over the life course ([Link and Phelan 1995](#); [Phelan and Link 2015](#); [Rothstein 2018](#)).

Owing to the role of segregation in perpetuating persistent Black-White inequalities associated, political efforts to promote racial equality have sought to promote integration across social space and institutional domains ranging from schools ([Johnson 2019](#)), hospitals ([Beardsley 1986](#)), and residence. Coinciding with the historical persistence of Black-White segregation is pronounced White resistance to residential integration through white flight ([Crowder and South 2008](#); [Frey 1979](#); [Logan, Zhang, and Oakley 2017](#)), avoidance from cities ([Rich n.d.](#)), and suburbanization ([Owens and Rich 2023](#)).

Established sociological explanations for the what drives Whites to resist progress toward racial equality is due to the perception of out group threats to their advantaged position ([Wetts and Willer 2018](#)). Drawing on the work of Blumer ([1958](#)), Bobo and Zubrinsky ([1996](#)) argue that resistance to racial integration can be seen as a form a response to “felt challenge,” ([Blumer 1958](#) : 5) wherein perceptions of out-group members becoming materially advantaged results in White backlash, especially in cases where there are material resources at stake ([Bobo and Zubrinsky 1996](#); [Quillian 1995](#)). While past work has evaluated the sources of

Whites' beliefs about threats to their status in the racial hierarchy, research has neglected to empirically examine whether there are material resources at stake or whether these reactions are primarily due to affect. Despite the fact that responses to integration suggest that Whites reap significant material advantages to segregation, the evidentiary basis for this is mixed. On one hand, effects of school desegregation shows that Whites children were largely unaffected and effects on White flight were muted ([Anstreicher, Fletcher, and Thompson 2022](#); [Johnson 2019](#); [Rich n.d.](#)). On the other hand, segregation has been shown to advantage Whites in meaningful ways, such as reducing their exposure to violent crime ([Light and Thomas 2019](#)). Segregation may also harm White Americans indirectly, by increasing their housing financial burden because of the cost-premium in majority white neighborhoods ([Daniels 1975](#)). What then, explains White resistance to integration?

I argue that we can characterize White resistance to integration in terms of perceived and material costs. Emerging from the sociological and political science literature on Whites responses to racial equality are the (1) **Zero-Sum** hypothesis and the (2) **Wages of Whiteness** Hypothesis. The unquestioned assumption in most scholarship is that White Americans necessarily benefit from residential segregation and that therefore, (1) integration would be harmful to the material advantages that they reap ([McGhee 2021](#)). However, an alternative view (2), advocated for by DuBois ([1935](#)) argues that White Americans are paid a psychological wage for their participation in White supremacy, despite the fact that they are in a neutral or similarly disadvantaged position as Black Americans.

Evidence on White Americans' public opinions on race and equality support the **Zero-Sum Perspective** that Whites' view progress toward racial equality, such as welfare expansion([Gilens 1996](#)), and policies more promoting equality negatively because of perceived threats to the status of Whites' position in the racial hierarchy ([McGhee 2021](#); [Wetts and Willer 2018](#)). In both potential scenarios, there are potential costs to

integration, however one is real and the other is imagined. Without understanding whether racial integration is materially disadvantaging to Whites' or primarily the result of incorrectly perceived threats to status, it is impossible to formally adjudicate which of these hypotheses explains the causal forces behind Whites' avoidance to integration.

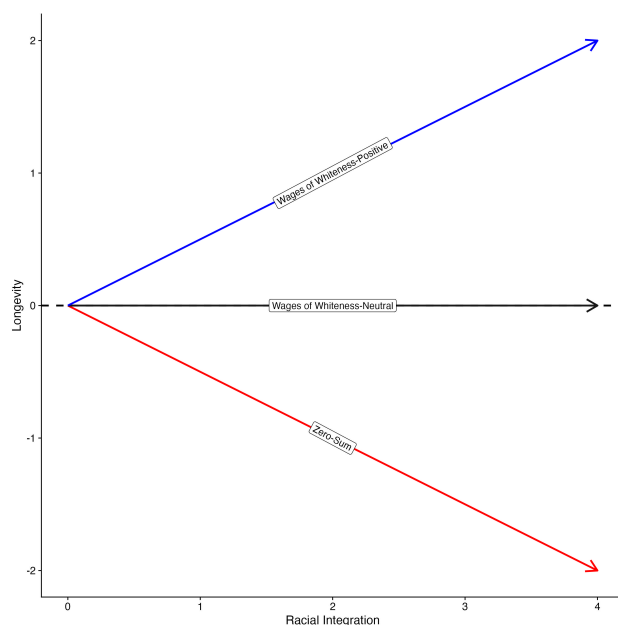


Figure 1: Theoretical Predictions of the Relationship Between Integration and White Longevity

To illustrate these potential explanation for White resistance to residential integration in the context of this study, Figure 1 describes three theoretical predictions generated from the **Zero-Sum** and **Wages of Whiteness** hypotheses that can then be validated with empirical analyses of the effects of segregation on White longevity. The first possibility, derived from a **Zero-Sum hypothesis** posits that increases in racial integration reduce White longevity advantages afforded by segregation. Beyond perceived threats to status position, higher levels of racial segregation result in material advantages for White Americans that would be reduced if residential spaces were more integrated.

In *Black Reconstruction in America*, DuBois (1935) emphasizes that the role of the psychological wage paid to Whites essentially offsets the poor material circumstances that Whites face by virtue of their similar class

position to Black Americans. This suggests that there is scope for Whites to be either unaffected or even disadvantaged by segregation. The second and third possibilities, derived from the **Wages of Whiteness Hypothesis** argue that increases in integration either do not affect White Americans or improve their longevity, the latter of which highlights the negative consequences of segregation that stem from its structural nature with system-wide implications.

3.2 Prior Work on Racial Segregation and Health

Prior work on segregation and outcomes has shown that segregation is associated with negative health outcomes of Black Americans. Using an instrumental variable strategy introduced by (?) leveraging exogenous variation in pre-settlement railroads, (?) found that segregation decreased the birthweight of Black children by 2.8 grams and decreased the birthweight of White children by .63 grams. Similarly, (?) also study effects on birthweight using the same instrument and find negative impacts of segregation on the birthweight of Black children. Their results indicate that the primary mechanisms that drive these outcomes arise from disparities in access to prenatal care for Black women, structural racism, and food insecurity. In contrast, (?) found that in some cases, segregation may be beneficial for Black mental health, though these processes may be distinct from those affecting physical health and longevity. Finally, the study that this paper is most closely related to is (?) that shows that segregation is responsible for a 16 year increase in the Black-White life expectancy gap for men and a 5 year increase for women, based on period life expectancy estimates between 1990-2019. This study makes important contributions to the empirical literature by exploring the effects of segregation on individual-level longevity that facilitate the understanding of how segregation affects health cumulatively and across the socioeconomic gradient.

At the same time, this work also has implications for White Americans. While the bulk of studies rightly focus on health outcomes for Black Americans, segregation is a system-wide phenomenon that transcends and

shapes the individual factors differentiating neighborhoods where Black and White Americans reside. (?) for instance found that there was no effect of segregation on the birthweight of non-Hispanic White children but (?) found that segregation decreased White childrens' birthweight significantly. Similarly, while (?) found that overall segregation widened life expectancy gaps between Black and White Americans, there were still county-level contexts where Black life expectancy was similar or exceeded that of White Americans.

While studies have documenting the effects of segregation on Black American's health and economic outcomes has been extensively documented, important methodological questions remain about whether these associations are causal and substantively, the processes through which segregation affects Black and White Americans. Research on the persistence of segregation over time draws attention to the complex ways in which individual preferences for neighborhood racial composition and structural sorting patterns affect where people live (?; ?). For instance, it could be that patterns of residential segregation and individual health and longevity are driven due to unobserved heterogeneity, preferences, or neighborhood characteristics like poverty and environmental externalities which result in the segregation-health relationship. Moreover, research shows that racial disparities in health persist at higher economic levels (?; ?) and that more advantaged Black populations often live in similarly segregated neighborhoods to poorer populations (?). Therefore, examining how effects vary by economic status contributes clarity about whether negative health outcomes for Black Americans are primarily the result of differences economic positions or the ability to convert resources to health for Black Americans or whether disparities are driven by structural racial inequality that transcends socioeconomic status.

3.3 Theory Linking Racial Segregation to Health

County-Level Mechanisms

In recent years, scholars have demonstrated that socioeconomic differences may not fully explain Black-White

health inequality because of Black Americans' residual negative health status that persists at higher SES levels (e.g., ?; ?; ?). Even as Black Americans ascend the economic ladder, they experience diminished health compared to White Americans (e.g., ?). One important driver of these disparities is structural racism, which systematically advantages White Americans and disadvantages Black Americans through interconnected social, legal, political, and education systems (e.g., ?).

To date, the majority of studies linking segregation to health outcomes have focused on between neighborhood mechanisms, but focusing on neighborhoods neglects important macro-level pathways that are consequential for health (e.g., ? 3; Montez et al. 2020). The role of administrative units, such as the county or state, are important from a public policy perspective, because they are intrinsically linked to important political, legal, and institutional processes that affect individuals living within these boundaries. Conceptually, focusing on the county-level as the unit of analysis is congruent with segregation theory and measurement, which refer to the distributions of populations at smaller geographies within a larger geography (e.g., ?). Studying segregation at smaller geographies may be sensitive to neighborhood change over time as census tracts change over time and their populations shift dynamically. Despite the fact that individuals may move between neighborhoods, the experience and effects of segregation are shaped by the different populations living in inter-dependent neighborhoods. While micro-level analyses capture how proximate neighborhood effects shaped by segregation affect individual outcomes, county-level analyses captures the average effect of segregation at a structural level irrespective of neighborhood differences.

At the county level, there are a variety of theoretically relevant and compelling pathways that may shape individual health outcomes. Figure 1, describes a stylized empirical diagram relating segregation to longevity, county-level and neighborhood pathways, and relevant confounding processes that are described in this section. Counties that are more segregated have are more likely to have decreased funding for public goods

and public opinion in these places indicates decreased support for redistributive policies that could boost the economic conditions of working class populations (McGhee 2021; ?). More generally (?), Gilens (2000), and McGhee (2021) hypothesize the common factor of these processes is greater racial resentment in more segregated counties. Through this mechanism, greater aversion to public policy that would likely improve the economic conditions of low-income Black Americans and decreased government support would likely have differential impacts on economic and material resources that have the potential to affect low-income populations more broadly, regardless of race.

[Figure 1 Here]

County-level mechanisms also work through the criminal legal system to affect individual outcomes through institutional and governmental mechanisms that are shaped by historical racial inequality (?, ?, ?). Compared to any other sex-age-race group, Black men and women are at the highest risk of experiencing police violence (?). (?) demonstrates that patterns of aggressive policing and punitive criminal justice policy contexts stem from high levels of segregation. In this way policing is tied to racial composition and the differing residential arrangements between White and Black Americans. Exposure to aggressive policing in segregated areas may plausibly affect both Black and White mortality through processes of arrest and incarceration. Additionally, Light and Thomas (2019) show that the segregation also creates racialized inequalities in homicide risk both through increasing Black risk, but also mechanically decreasing White Americans' risk of victimization because of residential separation. Beyond extreme outcomes like police use of force, exposure to aggressive policing leads to a higher likelihood of criminal justice contact, which has been linked to negative population health outcomes for Black Americans (?, ?).

County-level effects of segregation are also shaped by historical processes and long-run patterns of racism and stratification (Rothstein 2018; ?). (?) demonstrates that, in Northern metropolitan areas, segregation

led to an increase in poverty for Black Americans and decreased poverty for White Americans. Additionally, segregation increased Black-White income gaps. In contemporary contexts, (?) show that HOLC redlining maps, which were developed for the sole purpose of maintaining segregation, created greater economic isolation and devaluation of property values in mapped areas that have proximate implications for segregation and health outcomes through homeownership (?). At a broad level, research on the historical legacy of segregation suggests that even in an era of decline (?), patterns of segregation and racial inequality have reverberating effects over time that are important for present day outcomes.

Neighborhood-Level Mechanisms

County-level processes also have downstream effects on within-county neighborhood inequalities that shape health. In places with high levels of segregation, structural forces and individual residential mobility patterns shape residential decisions that are associated with social and environmental disadvantages and unequal opportunities across neighborhoods that stratify the neighborhood conditions of Black and White individuals (?).

Massey and Denton (2003) argue that segregation ultimately leads to greater concentrations of poverty, which is consequential for both economic mobility and health (?). Poverty affects health through mortality directly, but also through indirect pathways that influence how health status changes over the life course through exposure to poor residential conditions, resource deprivation, and increased exposure to illness. Moreover, the levels, timing, and duration of poverty are also consequential, particularly for health in adolescence (??; ??; ??). Segregation leads to higher spatial concentrations of poverty and income segregation within racial groups (?), that overlap with Black and White residential patterns. Racially stratified exposure to poverty has both direct effects on Black American's health outcomes by reducing educational and economic opportunities and also indirectly benefits White Americans' health by reducing their exposure (??; ?).

Environmental characteristics of neighborhoods are also an important driver of health (?) shaped by both exposure as well as ultimate political processes and public policies that allowed pollution in poor and Black areas (? 20). Black neighborhoods are frequently exposed to hazardous environmental conditions and among the highest risk of being exposed to air pollution (?). Black and low-income populations are also at higher risk of exposure to toxic chemicals and metals such as lead (?) which have direct and indirect effects on health and cognitive ability. More generally, these exposures reflect a geography of hazardous environmental exposures is racially and economically stratified, with polluting industries affecting both current sites of production as well as leaving behind remnants of emissions and toxins in disproportionately minority neighborhoods (? 6).

The neighborhood context also significantly shapes the educational opportunities and school quality of Black and White children ([Wodtke et al. 2023](#)) both through racially stratified education opportunities and school funding is tied to homeownership and property taxes. Black and White students also attend segregated schools that are shaped by the demographics of their neighborhoods and correlated with educational achievement over the life course (?). A prior study by (?) found that all else equal, segregation was responsible for a quarter of the Black-White gap in Scholastic scores.

Despite the strong associations between segregation and health, an important differentiator of Black-White health disparities is the role of educational disparities. For instance, Black Americans have on average lower education than White Americans, however, certain mechanisms that are hypothesized to stem from segregation may affect those with lower-education and economic status, regardless of race. A large body of research shows that attaining greater education is associated with better health and longevity at an individual and population level, making it complicated to separate the effects of one's environment on their health from individual characteristics ([Link and Phelan 1995](#); ?; ?). Education is an important aspect of upward mobility and social status because it is a valuable resource that accumulates over time and shapes individuals' health

through knowledge, behaviors, expansion of economic opportunities, and access to health promoting resource like insurance and healthcare (?). Education also increases individuals ability to convert flexible resources into health promoting and protecting behaviors (?). Those who attend schools with greater resources and funding frequently concentrated within White neighborhoods, benefit from the racially stratified geography of educational opportunity.

Because of the highly coupled relationship between segregation, neighborhood opportunity, and educational attainment, it is possible that the relationship between segregation and health could be driven by underlying educational disparities. Furthermore, even if segregation primarily shapes Black and White longevity, education may moderate this relationship in important ways.

3.4 The 1940 Context

There are a few features of the data used in this study as well as the observed period of segregation that are important to contextualize when interpreting estimates in this study. The birth cohorts used in this study from 1905-19xx survived exceptional economic hardship from the Great Depression as well as the 1918 Flu Pandemic and were members of the earliest cohorts of Black migrants from the South to the North. These circumstances reflect important selectivity that must be taken into account when situating these estimates within the literature on segregation and health.

The purpose of using the 1940 Census data for analysis are to identify individuals who were part of the 1904-1918 birth cohorts who likely complete their mortality window within the 1988-2005 time frame that the Social Security Administration public data is available for (?; ?). These birth cohorts are also socially and historically significant. Members of these birth cohorts are situated in a historical period marked by intense hardship followed by prosperity. These cohorts survived the first and second World War as well as the 1918 Spanish Flu which also had distinct racial disparities (?). During the 1920s and 1930s, these birth

cohorts weathered the great depression, where unemployment reached highs of 24 percent (?). Amid these major changes to American Life were the prosperity that followed these material hardships, spurred in large part by expanded opportunity resulting from the New Deal introduced by President Franklin D. Roosevelt. At the same time as these changes happened, the Great Migration of Black Americans from the South to the North that occurred between 1910 and 1970 led to a restructuring of the landscape of opportunity for Black Americans in the U.S. as reshaped race relations in complex ways (?).

The time period of life that I observed segregation is also important to consider when interpreting estimates. I examine exposure to segregation that mid-to-late life, obtained and calculated based on the county individuals resided at death. This decision is to maximize geographic coverage within and outside the South to better reflect contemporary distributions of Black and White Americans. Prior to the great migration [XXXX percent of Black Americans lived in the North].

Research on the life-course dynamics of migration propensity (?) suggests that individuals are less likely to migrate after age 40 years old and when they do it is usually short distances (?; ?). Additionally, research on aging in place shows that older adults increasingly reside in the same neighborhoods and residential contexts for 30 or more years preceding death (?). For these reasons, examining exposure to segregation in the county of death represents a wider window of exposure stretching back to mid-life. It is also true that prior to the period of segregation I observe, average segregation levels of segregation in counties in my analytic sample were higher than the observation period. Coupled with this trend, work on neighborhood attainment and residential mobility show that the modal neighborhood transition is to one of similar characteristics (?). Thus, I conclude that estimates of exposure to segregation experienced in mid-to-late life, net of birth place characteristics, proxy for life course segregation exposure that are likely attenuated from the levels experienced earlier in life.

In sum, taking selective characteristics into account, results should be interpreted as conservative estimates of segregation on longevity that are likely attenuated from what their true magnitude.

3.5 The Present Study

The goal of this study is estimate the causal effect of segregation on longevity. Longevity, defined as the number of years one lives conditional on surviving to 65 years old, is an important omnibus measure of health and life chances (Weber 1922; ?). To address this question, I combine linked data from the 1940 Census with data on municipal and township governments to instrument for residential segregation to identify a causal effect of segregation on longevity. With these empirical and theoretical considerations in mind, the present study addresses two research questions.

- (1) What is the effect of county-level racial segregation on Black longevity?
- (2) What is the effect of county-level racial segregation on White longevity?
- (3) How do the effects of segregation on longevity vary by educational status for Black and White Americans?

4 Data

4.1 Data Sources

To answer my research questions, I leverage data on 1904-1924 birth cohorts of persons xx-xx years old from the 1940 Full Count United States Census. I link these data with Numident death records and the Supplementary Geographic File provided by the Berkeley CenSoc project that combines geographically linked death records from the Social Security Administration (SSA) (?).

The geographic data in the Numident Geographic Supplement cover deaths occurring between 1988-2005 and reliably cover deaths over the age of 65 during this period. The geographic specificity and rich covariates,

harmonized over time, make these ideal for analyses of longevity. I focus on the 1904-1924 birth cohorts for two reasons.

First, their birth years and the end year of the mortality observation window allow for likely complete observation of mortality that is not biased by missing the oldest old. In other words, those born in 1904 would be 101 years old in 2005, making it safe to assume that there is not substantial missing, right-censored data on individuals who are still alive. Second, those aged 22-36 likely have completed the majority of their formal education, which allows me to rule out the possibility of confounding related to individuals who are still in the process of completing education (?).

My final analytic sample consists of ($N = 3,027,772$) individuals residing in 1,809 – *check* counties at their time of death. These counties are then linked to 1967 Census of Governments data on the number of municipal and township governments and their Federal Government revenue share from the Government Finance Database (**CITE**) that I use for identification. The focal filtering criteria that I use is coverage of my instrument, and Table (xxxx) describes the differences between non-sample versus sample counties, that appear to be broadly representative.

To measure segregation, I use the dissimilarity index, a measure that describes how evenly Black and White residents are distributed in sub-geographies within a county relative to the county-level racial composition, which has been used widely in empirical research (?). As mentioned previously, I calculate the dissimilarity index for the county an individual resided in at the time of their death. The dissimilarity index is interpreted as the share of Black and White individuals that would have to move for the racial composition of neighborhoods to be the same as the racial composition of the county. The dissimilarity index is calculated formally as:

$$D = \frac{1}{2} \sum_i^N \left| \frac{b_i}{B} - \frac{w_i}{W} \right| \times 100$$

Where N is the number of counties, b_i and w_i are the proportion of Black and White residents in the i th census tract and B and W are the proportion of Black and White residents in the county. The 1940 Census also includes data on county of residence at birth which I compare to county at death to determine whether individuals were movers or stayers. The dissimilarities are calculated from decennial Censuses 1980, 1990 and 2000, which correspond to the year that they passed.

Covariates and Adjustment for Double Truncation

In addition to the instrumental variable strategy, I control for a rich set of individual and contextual covariates to adjust for any residual confounding. In fully specified models, I control for sex, education, marital status, employment status, whether an individual migrated, whether they reside in the South, urban-rural fixed effects, birth year and state fixed effects, period fixed effects, and occupation fixed effects.

Birth year fixed effects are integral to reducing bias in mortality analysis with truncated data, where individuals of varying birth cohorts may have differing numbers of observations covered by the death window.

It is well known that the NUMIDENT data suffer from the problem of double truncation that refers to the left and right censoring of the data (?). (?) show that when not accounted for, doubly truncated observation windows result in biased estimates of longevity disparities toward 0 and mean age at death is higher. This means that, for those born in 1904, we observe them between the ages of 84-101 years old and for those born in 1922 we observe them between the ages of 70-87 years old.

It is also possible that there is important unobserved heterogeneity in early life experiences and residential environments common to one's place of birth that could bias estimates of segregation on longevity. To adjust

for early life characteristics of individual birth places, I include birth county fixed effects from the geographic supplement in the NUMIDENT to address early life exposures.

Despite the strengths of these data there are important limitations. In terms of record matching algorithms, matches between the 1940 Census and the NUMIDENT may favor more advantaged Black Americans, resulting in estimates of the effect of segregation that are conservative. These analyses are also conditional on having no missing data in the NUMIDENT geographic supplement, which adds an additional layer of selectivity. Substantively, doubly truncated data is also a limitation because, while it is possible to recover estimates of longevity disparities, the narrow time window limits understanding whether longevity outcomes are primarily caused by earlier or later exposures to segregation (?). To address these representation issues, I use post-stratification weights developed by (?) that re-weight CenSoc NUMIDENT counts to year-specific distributions of deaths to adjust for possible selection. Table 7 in the appendix compares the analytic sample used in this paper to the full NUMIDENT population and these populations are, on average, comparable though my analytic sample is positively educationally selected and live in slightly more segregated counties.

5 Methods

5.1 Identification Strategy

Research has established that there are significant barriers to drawing causal inferences from observational data on the relationship between segregation and longevity. In particular, recent work on the place-level correlates of population health disparities show that there are many features of state-level policies contexts, which necessarily affect counties, that adversely shape health (Montez et al. 2020), leading simple associations between county-level segregation and longevity to be biased.

A leading strategy to alleviate concerns of endogeneity and unobserved heterogeneity is to find an instrumental

variable that induces exogenous variation in segregation, but does not plausibly affect the outcome of interest, except through segregation. To identify a causal effect of segregation on longevity, I use an instrumental variable strategy first developed by Cutler and Glaeser (1997) and employed subsequently by Light and Thomas (2019) that leverages variation in the number of municipal and township governments and the state-level federal government revenue share these governments receive as instruments.

These instruments are thought to induce segregation through a Tiebout mechanism: “when there are more local governments, tax rates and service provision will vary more within an area, and thus the desire for sorting will increase,” (Cutler and Glaeser 1997:848). This quasi-randomly induced sorting is thought to mechanically lead to more segregation that is not related to empirical outcomes of interests (e.g., income, test scores, crime). In this paper, I extend this strategy to study health and longevity. While it may be the case that the number of governments and quality of services typical to each government may be correlated with health, I rely on the 1967 Census of Governments which is sufficiently temporally prior to the study period of 1980-2000. As noted by Cutler and Glaeser (1997) and Light and Thomas (2019), the correlation in the number of governments over time is very high at over .98 (Cutler and Glaeser 1997:848). Furthermore, these government patterns exhibit a high degree of correlation across an even longer time horizon, stretching back to 1787 in some cases, suggesting that their influence on sorting patterns occurred sufficiently prior to the segregation studied in this paper. For the purposes of this paper, information from the Census of Governments are obtained from the Government Finance Database (cite) and merged to counties to obtain aggregated information. In the regression setting, I follow Cutler and Glaeser (1997) and include the log of the number of governments and the state-level share of intergovernmental revenue to purge local endogenous factors.¹

When evaluating the plausibility and relevance of this instrument, it is important to do so in the context of

¹The first stage is estimated as $D = \alpha + \ln(n_governments) + revenue_share + \epsilon$.

IV assumptions. (1) the instrument is exogenous and relevant (this cannot be tested formally), (2) strength of the first stage relationship (the IV must substantially affect the endogenous variable), and (3) monotonicity (the IV only influences the level of segregation in a county in one direction).

First, the assumption that (1) these instruments impact longevity only through segregation. It is possible that could affect other population outcomes (e.g., racial composition or population size) that may affect longevity and segregation through unobserved pathways. Figure X indicates that the association between the instruments and population characteristics support the conclusion that these instruments did not affect population characteristics that are components of segregation (HMM think about this). (2) In Figure X and table X, I demonstrate that association between the instruments and segregation is highly predictive. The first-stage F-statistic tests that test the strength of the instrument and segregation and indicate a strong and significant F-test of **XXXX** (Table 2 in Appendix) (?). and (3) it is unlikely that these instruments result in sorting that reduced the level of segregation in a city [**This merits further discussion**]

Because there are more instruments than endogenous regressors, I conduct a Sargan overidentification test that checks whether the instruments are appropriate and identify the same parameters [**Get Cite**], and this test cannot reject the hypothesis that that the instruments should be included ($p = .xx$).

Provided that these assumptions are satisfied, IV estimates of a regression of longevity on segregation are interpreted as the local average treatment effect (LATE) (?), or the treatment effect amongst those living in counties where segregation increased as a result of greater sorting from increased government density Despite limitations, to date, these instruments has been used effectively in prior studies to examine how segregation effects multiple outcomes ([Cutler and Glaeser 1997](#); [Light and Thomas 2019](#)) In the context of trying to recover estimates as close to causal as possible, randomized experiments manipulating individual residential contexts are not feasible nor ethical.

5.2 Estimation

Empirically, I estimate the relationship between segregation and age at death, conditional on living to 65, in the county of residence at death. I estimate the following regression via two-stage least squares estimation, stratified by racial group (?). In all models include robust standard errors to account for heteroskedasticity. Standard errors clustered at the county-decade level to account for spatial and temporal variation yield similar results.

$$Y_i = \beta_0 + \beta_{Segregation} + \gamma_{birth-year} + \lambda_{birth-county} + \beta X_{controls} + \epsilon_i$$

In summary, this empirical strategy addresses critical endogeneity and selection issues in estimating the relationship between segregation and longevity, stemming from individual preferences and structural constraints and confounding the relationship. Based on the assumption that the railroad division only influences longevity through segregation, estimates can be interpreted as causal. Moreover, interpreting these results in context, the nature of the selectivity of the 1904-1922 birth cohorts, the selectivity of examining life expectancy over 65, and the selectivity in how individuals are matched from the 1940 Census to NUMIDENT point toward these estimates being conservative and attenuated. For Black Americans, in particular, descriptive statistics indicate that this sample is advantaged along death age and education, suggesting that effects of segregation are also likely attenuated from what the total population effects would be. In an ideal world with complete data coverage, estimates of the effect of segregation on longevity would likely be higher than the results presented in this paper.

6 Results

7 Discussion

Answering this question thus helps clarify the roots of White aversion to integration but also enriches scholarly understanding of the distributional consequences of racial inequality.

Understanding of the material consequences of segregation for Whites is important because it otherwise limits our ability to understand the distributional consequences of racial inequality and how they shape actors' decisions in support or opposition.

More generally these results limit our scholarly understanding on the distributional consequences of segregation. While subtle, establishing this distinction is particularly important for making policy to redress profound racial inequalities in health and longevity, because effective policies require broad appeal including a significant share of the voting bloc, which is White.

which has implications for policy aimed at redressing racial inequalities in health.

a zero-sum hypothesis argues that segregation materially advantages Whites, a Du Boisian hypothesis argues that segregation affords psychological advantages despite neutral or even negative effects of racial inequality for Whites.

8 Tables and Figures

References

Anstreicher, Garrett, Jason Fletcher, and Owen Thompson. 2022. "The Long Run Impacts of Court-Ordered Desegregation."

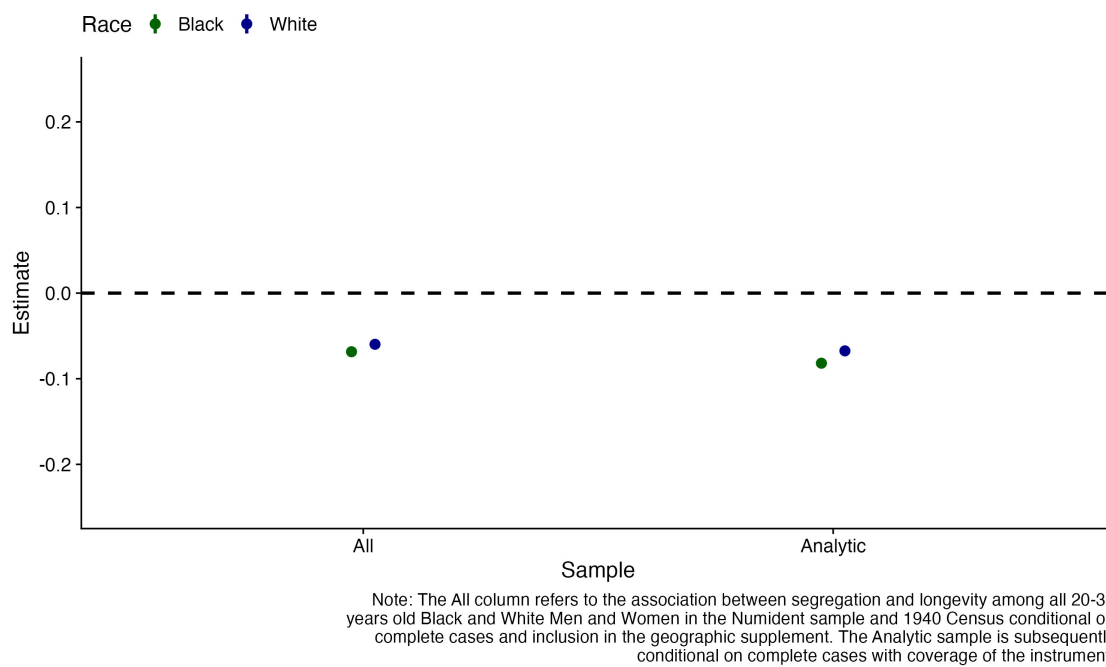


Figure 2: Association Between Seg and Longevity by Sample

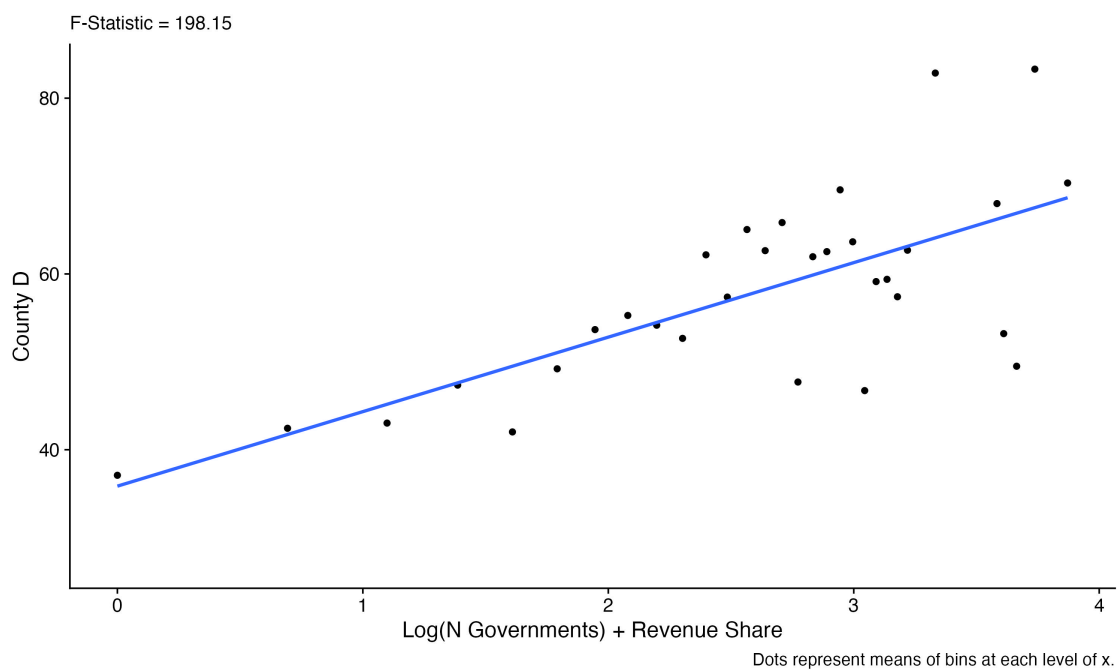


Figure 3: First Stage

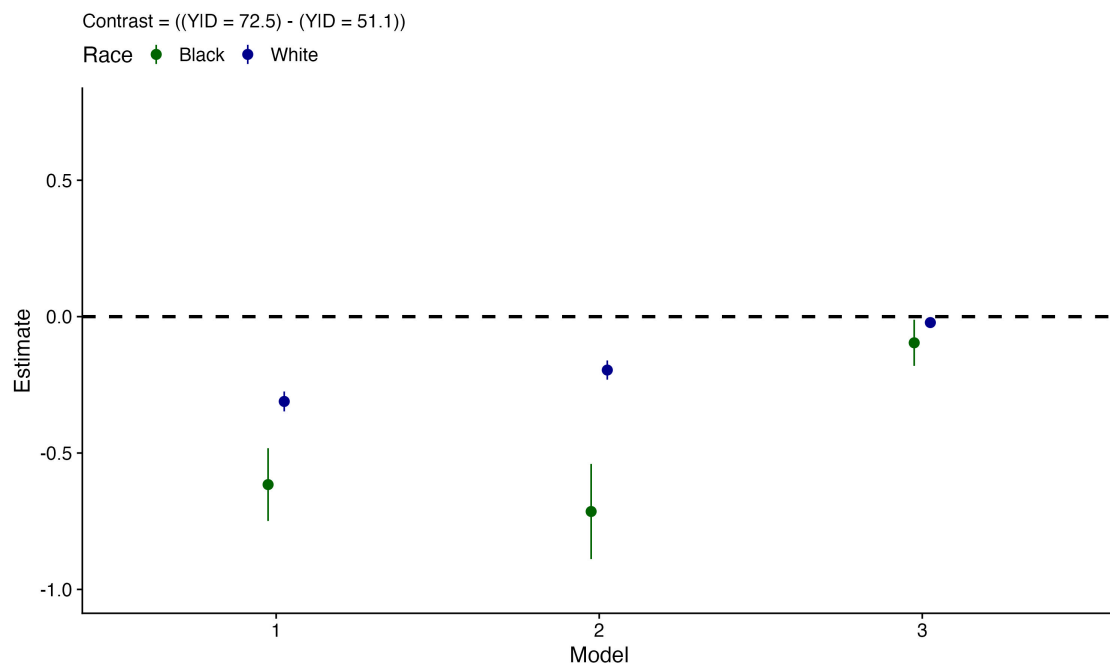


Figure 4:

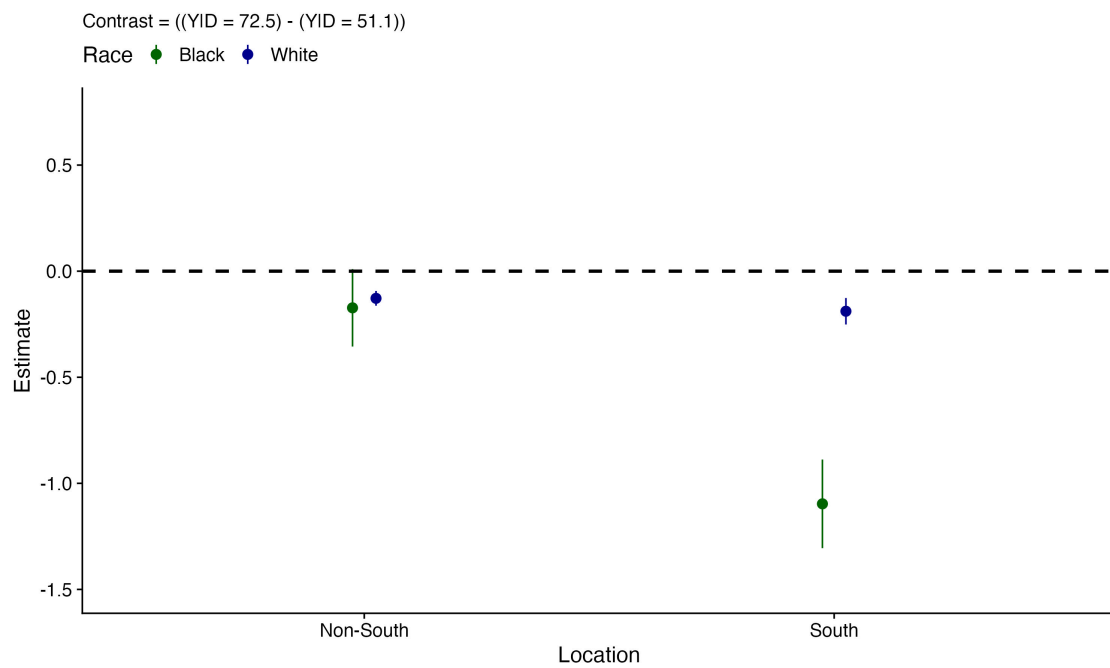


Figure 5:

Table 1: First Stage Regression of D on Instruments

	(1)
Ln(Number Governments)	7.45*** (0.29)
County Share Revenue from Federal Gov.	25.60*** (6.06)
Num.Obs.	4855
F	198.15

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001
First stage relationship includes death decade fixed effects and heteroskedasticity SEs.

Table 2: Estimates of the Association Between Segregation and Longevity

	Black	Black + C	Black + C + Period	White	White + C	White + C + Period
D	-0.082*** (0.001)	-0.084*** (0.001)	-0.004*** (0.001)	-0.067*** (0.000)	-0.067*** (0.000)	-0.001*** (0.000)
Male	-1.055*** (0.029)	-1.008*** (0.030)	-0.209*** (0.021)	-0.935*** (0.007)	-0.944*** (0.007)	-0.213*** (0.005)
Education		0.048*** (0.004)	0.007** (0.002)		0.062*** (0.001)	0.013*** (0.001)
Migrated		-0.455*** (0.037)	-0.068*** (0.018)		0.016* (0.008)	-0.092*** (0.004)
Married in 1940		0.088** (0.031)	0.073*** (0.016)		-0.059*** (0.007)	0.022*** (0.004)
Num.Obs.	108681	108681	108681	2048843	2048843	2048843
R2	0.387	0.389	0.858	0.350	0.352	0.849

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001
Heteroskedasticity Robust Standard Errors in parentheses.

Table 3: Estimates of the Effect of Segregation on Longevity

	Black	Black + C	White	White + C
D	-0.029*** (0.003)	-0.033*** (0.004)	-0.015*** (0.001)	-0.009*** (0.001)
Male		-1.009*** (0.045)		-0.955*** (0.010)
Education		0.037*** (0.005)		0.061*** (0.001)
Migrated		-0.256*** (0.039)		-0.209*** (0.008)
Married in 1940		0.127*** (0.033)		-0.017* (0.008)
Died in South		0.062 (0.055)		0.124*** (0.010)
Num.Obs.	108681	108681	2048843	2048843
R2	0.367	0.379	0.327	0.332
Birth Year	Yes	Yes	Yes	Yes
Birth State	Yes	Yes	Yes	Yes
Urban Rural Code	Yes	Yes	Yes	Yes
OCC	No	Yes	No	Yes

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$
Heteroskedasticity Robust Standard Errors in parentheses.

Table 4: Estimates of the Effect of Segregation on Longevity (By Gender)

	Black (Men)	Black (Women)	White (Men)	White (Women)
D	-0.033*** (0.006)	-0.032*** (0.006)	-0.011*** (0.001)	-0.008*** (0.001)
Education	0.056*** (0.008)	0.024*** (0.006)	0.083*** (0.002)	0.044*** (0.002)
Migrated	-0.251*** (0.061)	-0.261*** (0.050)	-0.163*** (0.013)	-0.246*** (0.011)
Married in 1940	0.121* (0.053)	0.134** (0.044)	-0.078*** (0.012)	0.061*** (0.012)
Died in South	0.004 (0.081)	0.125+ (0.074)	0.129*** (0.015)	0.118*** (0.013)
Num.Obs.	46166	62515	893437	1155406
R2	0.336	0.386	0.256	0.350
Birth Year	Yes	Yes	Yes	Yes
Birth State	Yes	Yes	Yes	Yes
Urban Rural Code	Yes	Yes	Yes	Yes
OCC	Yes	Yes	Yes	Yes

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Heteroskedasticity Robust Standard Errors in parentheses.

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Table 5: Estimates of the Effect of Segregation on Longevity (Weights)

	Black	Black + C	White	White + C
D	-0.032*** (0.004)	-0.035*** (0.005)	-0.016*** (0.001)	-0.012*** (0.001)
Male		-1.144*** (0.051)		-1.094*** (0.012)
Education		0.036*** (0.006)		0.063*** (0.002)
Migrated		-0.238*** (0.045)		-0.217*** (0.010)
Married in 1940		0.109** (0.037)		0.025* (0.010)
Died in South		0.086 (0.062)		0.108*** (0.012)
Num.Obs.	108681	108681	2048843	2048843
R2	0.383	0.396	0.366	0.373
Birth Year	Yes	Yes	Yes	Yes
Birth State	Yes	Yes	Yes	Yes
Urban Rural Code	Yes	Yes	Yes	Yes
OCC	No	Yes	No	Yes

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Heteroskedasticity Robust Standard Errors in parentheses.

Table 6: Estimates of the Effect of Segregation on Longevity (Stayers)

	Black	White
D	-0.027*** (0.004)	-0.010*** (0.001)
Male	-1.008*** (0.051)	-0.951*** (0.011)
Education	0.040*** (0.006)	0.065*** (0.001)
Married in 1940	0.075* (0.038)	-0.049*** (0.010)
Died in South	-0.002 (0.049)	-0.023** (0.009)
Num.Obs.	82083	1444492
R2	0.374	0.337
Birth Year	Yes	Yes
Birth State	Yes	Yes
Urban Rural Code	Yes	Yes
OCC	Yes	Yes

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Heteroskedasticity Robust Standard Errors in parentheses. Estimates reflect the subsample of individuals who were born and died in the same county.

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